AlphaBench — Technical Report

> **Disclaimer.** AlphaBench is an academic research and software-engineering artifact. It is not investment advice, not a trading product, and must not be used to make financial decisions. All results below are historical simulations and carry no guarantee of future performance.

1. Question and methodology

Question. Given a universe of liquid equities and their daily OHLCV history, can a supervised model predict the sign of the forward *h*-day return with accuracy statistically distinguishable from chance, and does that edge survive realistic transaction costs in a walk-forward backtest?

Target reframing. The naive formulation — predict tomorrow's *price* — is broken for three reasons: prices are non-stationary (a model fit on ₹500 stocks won't transfer to ₹3,000 ones), the series is highly autocorrelated (`Close[t+1] ≈ Close[t]`, so a persistence forecast trivially achieves low RMSE that looks impressive and means nothing), and no real decision depends on the price level, only on the change. AlphaBench predicts direction instead: $r_{t+h} = \log(C_{t+h}/C_t)$, labelled with a volatility-scaled deadband ($\kappa = 0.3 \times \sigma_t$, σ_t a 20-day trailing realised volatility computed strictly from data up to t) so the model is not forced to take a position on days that are pure noise.

Feature set. ~60 features (47 after excluding OHLCV pass-through columns) across six groups — momentum/reversal, volatility, technical indicators, volume, cross-sectional ranks (only possible because this is a 30-ticker panel, not a single name), and market/calendar — every one lag-shifted by one day within its symbol before it can reach a model (`features/base.py::safe_shift`), enforced structurally rather than by convention.

Validation. Expanding-window walk-forward cross-validation with purging (drop training rows whose label window overlaps the validation block) and a 5-day embargo (additional gap to break serial correlation at the boundary) — `validation/splitters.py`. Seven folds, validating 2017 through 2023 in turn on an expanding 2010-start training window. `tests/test_leakage.py` asserts train always strictly precedes validation, the purge/embargo gap is honoured, no feature correlates with the forward return above 0.10 on a synthetic random walk, and shuffled labels collapse AUC to ~0.50 — the harness itself is calibrated, not just the model. A final holdout block (2025-01-01 onward) was frozen from the start and touched exactly once, at the very end (section 6).

Backtest. Signal from day t's close, execution at day $t+1$'s open (the OOF probability is already aligned this way), equal-weight positions, 5 bps commission + 5 bps slippage per side applied to turnover, no leverage, no shorting. Position fires only when the model's probability clears a threshold calibrated to that model's *own* out-of-fold probability distribution (its 90th percentile — necessary because raw probabilities cluster tightly around 0.5 at this signal-to-noise level; not a search for the threshold that maximises backtest P&L).

2. The baseline ladder — B0 first

Primary market: NSE, 30 large-cap Indian equities (29 resolved), h=1 day, mean across the 7 walk-forward folds (2017-2023):

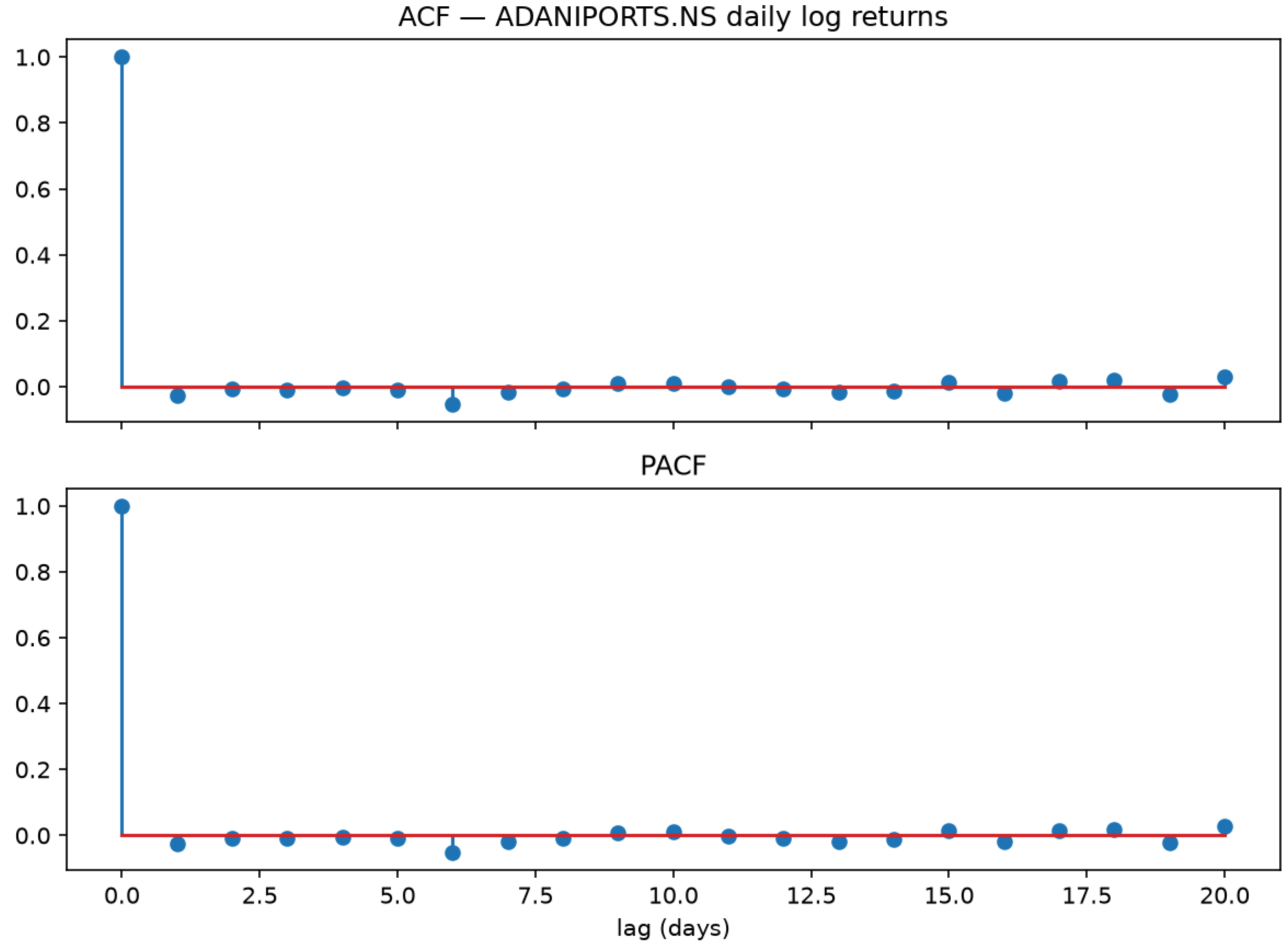
Model	ROC-AUC	Accuracy	Base rate
B0 persistence (today's direction = tomorrow's)	0.487	0.488	0.518
B0 majority class	0.500	0.518	0.518
B1 logistic regression (5 features)	0.510	0.518	0.518
B2 ARIMA(1,0,1) on log returns	0.503	0.511	0.518
M1 LightGBM (primary)	0.506	0.518	0.518
M2 XGBoost	0.515	0.518	0.518
M3 LSTM (PyTorch, CPU, small)	0.515	0.518	0.518
M4 rank-average ensemble (M1+M2+M3)	0.514	0.508	0.518

Full per-fold detail: `reports/metrics/baseline_results_h1.json`, `walkforward_results.json`, `walkforward_results_{arima,xgboost,lstm,ensemble}_h1.json`.

Every entry sits in a 0.487-0.515 band, and each model's own fold-to-fold standard deviation is roughly 0.01-0.02 (e.g. M1: 0.506 ± 0.020; M3: 0.515 ± 0.019 — see the per-fold JSON). **The gaps between models are smaller than the noise within any one model.** There is no ladder to climb here in any statistically meaningful sense — B0 majority (AUC exactly 0.500 by construction) is not reliably beaten by anything.

B2 — ARIMA diagnostics

ADF and KPSS were run on every symbol's own daily log-return series (never on price levels — `reports/metrics/arima_diagnostics.json`). Both tests agree the series is stationary for **29/29 symbols** (ADF rejects the unit-root null; KPSS fails to reject stationarity), which is the textbook result for daily returns and justifies fitting ARIMA directly with $d=0$. The ACF/PACF of a representative symbol (`reports/figures/arima_acf_pacf.png`) is indistinguishable from white noise beyond lag 0 — there is essentially no linear autocorrelation structure for any ARMA order to exploit, which is consistent with B2's AUC landing at 0.503, statistically indistinguishable from B0 majority's 0.500.



ACF and PACF of daily log returns for a representative symbol. Both are flat beyond lag 0 — there is essentially no linear autocorrelation structure for any ARMA order to exploit.

M3 — the LSTM comparison, reported honestly

`docs/PROPOSAL.md` section 4.2 predicted the LSTM would lose to the GBM at this data scale — the standard, well-supported expectation for ~114k rows of low-SNR tabular-shaped data. It did not lose here: M3 (0.515 ± 0.019) came in essentially tied with M2 XGBoost and slightly *above* M1 LightGBM (0.506 ± 0.020). Two things are true at once: this is reported as it happened rather than adjusted to match the prediction, and it does not actually contradict the prediction in any way that matters — every model's confidence interval overlaps every other model's, so "M3 beat M1" is not a claim this sample size can support. The honest statement is that no model in the ladder is distinguishable from any other, LSTM included.

The Prophet rejection

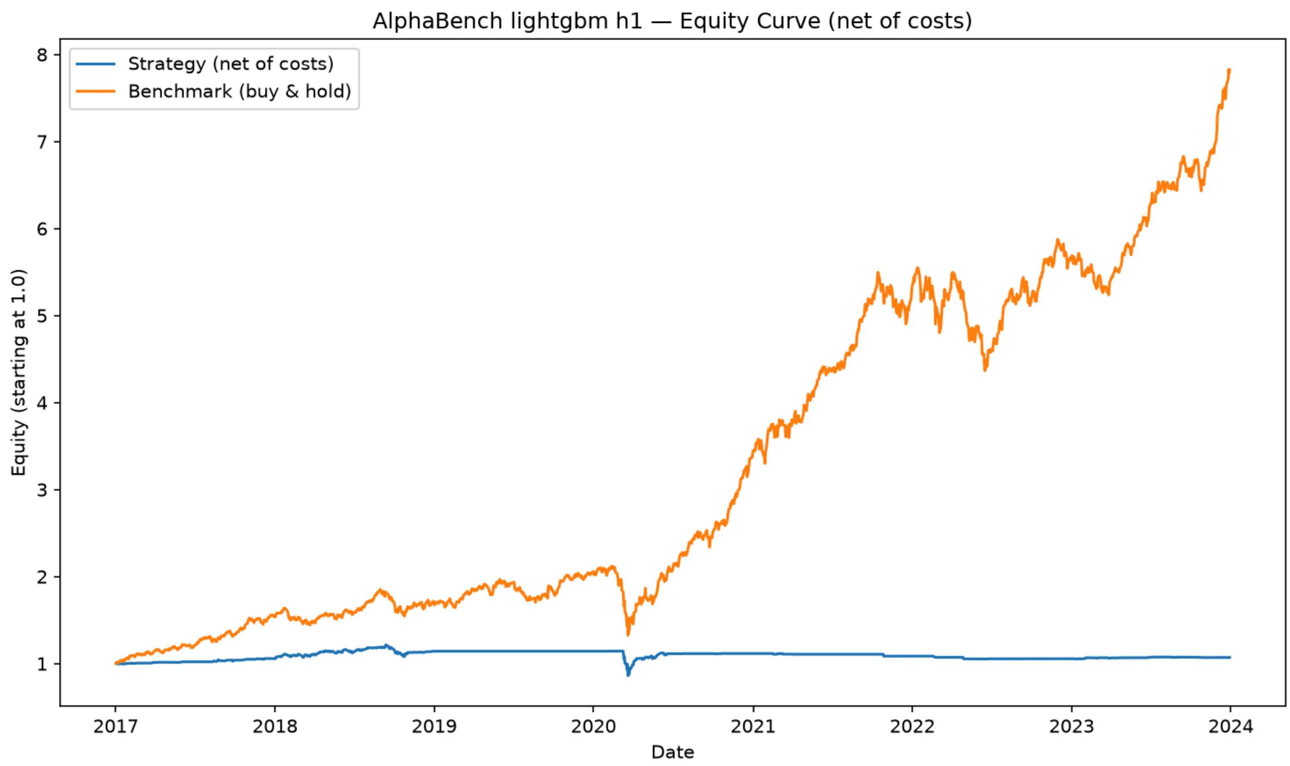
Prophet decomposes a series into trend, seasonality, and holiday components — built for business time series with real calendar structure (retail demand curves, web traffic cycles, holiday spikes). Daily equity log returns have none of that: the ACF/PACF above shows no periodic structure to decompose, and the mean daily return is close enough to zero that "trend" is not a meaningful concept over a 1-day horizon. Fitting Prophet here would mean handing it noise and asking it to find smooth curves in it — which it will do, convincingly, because that is what a smoothing model does regardless of whether structure is actually present. That combination (a model guaranteed to produce plausible-looking output on data it has no business modelling) is worse than not including it: it invites false confidence rather than testing a real hypothesis. B2 ARIMA already covers the "classical statistical baseline" role properly, on a specification (autoregression on returns) that at least matches the data-generating process being examined.

3. Economic evaluation

Headline (NSE, M1 LightGBM, net of 10 bps)

Metric	Strategy	Buy-and-hold NIFTY
Sharpe (annualised)	**0.108**	1.797
Ann. return	1.00%	34.96%
Max drawdown	−29.1%	−37.5%

Excess Sharpe: **−1.69**. The strategy does not beat buy-and-hold; it substantially underperforms it, both before and after accounting for the benchmark's own larger drawdown.



NSE h=1 equity curve, net of 10 bps, against equal-weight buy-and-hold on the same axes. The strategy is the flat line; the benchmark is the one that compounds.

Cost-sensitivity sweep (NSE)

Total bps	Sharpe	Ann. return
0	0.353	3.27%
2	0.303	2.81%
5	0.230	2.13%
10	**0.108**	**1.00%**
20	−0.131	−1.22%

The edge is not robust even before costs (Sharpe 0.35 gross is modest on its own), and it crosses zero between 10 and 20 bps — a realistic commission+slippage assumption is enough to erase it. Full table: `reports/metrics/backtest_results.json`.

Bootstrap confidence intervals

- **Sharpe** (block bootstrap, 21-day blocks, 2000 resamples): 0.155, 95% CI **[−0.45, 1.13]**, P(Sharpe > 0) = 0.699. The interval comfortably contains zero.
- **Holdout AUC** (block bootstrap over dates, 2000 resamples): 0.505, 95% CI **[0.482, 0.527]** (section 6). Straddles 0.50.

Deflated Sharpe ratio

A 50-trial Optuna search over LightGBM hyperparameters (`reports/metrics/optuna_study_h1.json`) found a best mean-CV-AUC of 0.5222 (vs. 0.5060 for the shipped `DEFAULT_PARAMS` model — the model that is actually reported everywhere in this document; the tuned parameters were never used to retrain the shipped model, precisely to avoid reporting a result inflated by that search). Correcting the reported backtest Sharpe (0.108) for the 50 trials run and the sample's own variability:

	Value
Observed Sharpe (annualised)	0.108
Null bar under 50 trials (annualised)	0.146
P(skill is real)	**0.46**
Significant at 95%?	**No**

The null bar — the Sharpe you would expect from the best of 50 random configurations with zero real skill — is *higher* than the Sharpe actually observed. Even without invoking the holdout, the walk-forward backtest result alone does not clear the bar a reviewer should expect from chance plus search effort.

Per-year breakdown (NSE)

Year	Ann. return	Sharpe
2017	+5.7%	1.89
2018	+8.0%	0.85
2019	−0.0%	−1.02
2020	−2.3%	0.01
2021	−2.7%	−1.25
2022	−2.9%	−1.45
2023	+1.5%	1.18

Sign flips essentially every other year with no visible regime pattern (not, for example, concentrated in the 2020 COVID crash) — the strategy has not found one event, it has found noise that occasionally points the right way.

Per-ticker breakdown (NSE)

Full table: `reports/metrics/backtest_results.json` (`by_ticker`). Per-symbol Sharpe ranges from **+0.62 (DRREDDY.NS)** to **−0.80 (NTPC.NS)**, with roughly half the 29 symbols positive and half negative. This directly answers `docs/PROPOSAL.md` section 7.3's question — "does the edge generalise, or is it two names carrying twenty-eight?" — and the answer is neither: there is no small subset of names driving the aggregate result: the aggregate itself is unremarkable, and the dispersion across names is exactly what you would expect from independent noise around a near-zero mean, not a real cross-sectional signal that happens to concentrate somewhere.

h=5 horizon

`config.yaml` declares `horizons: [1, 5]` ; h=1 is the primary result reported throughout this document, and h=5 (5-day-ahead direction, same LightGBM architecture and hyperparameters, same walk-forward folds) was trained and backtested identically for completeness.

Walk-forward: mean AUC 0.5210 ± 0.0283 across 7 folds — the same noise-level result as h=1 (0.506 ± 0.020), if anything slightly noisier fold-to-fold.

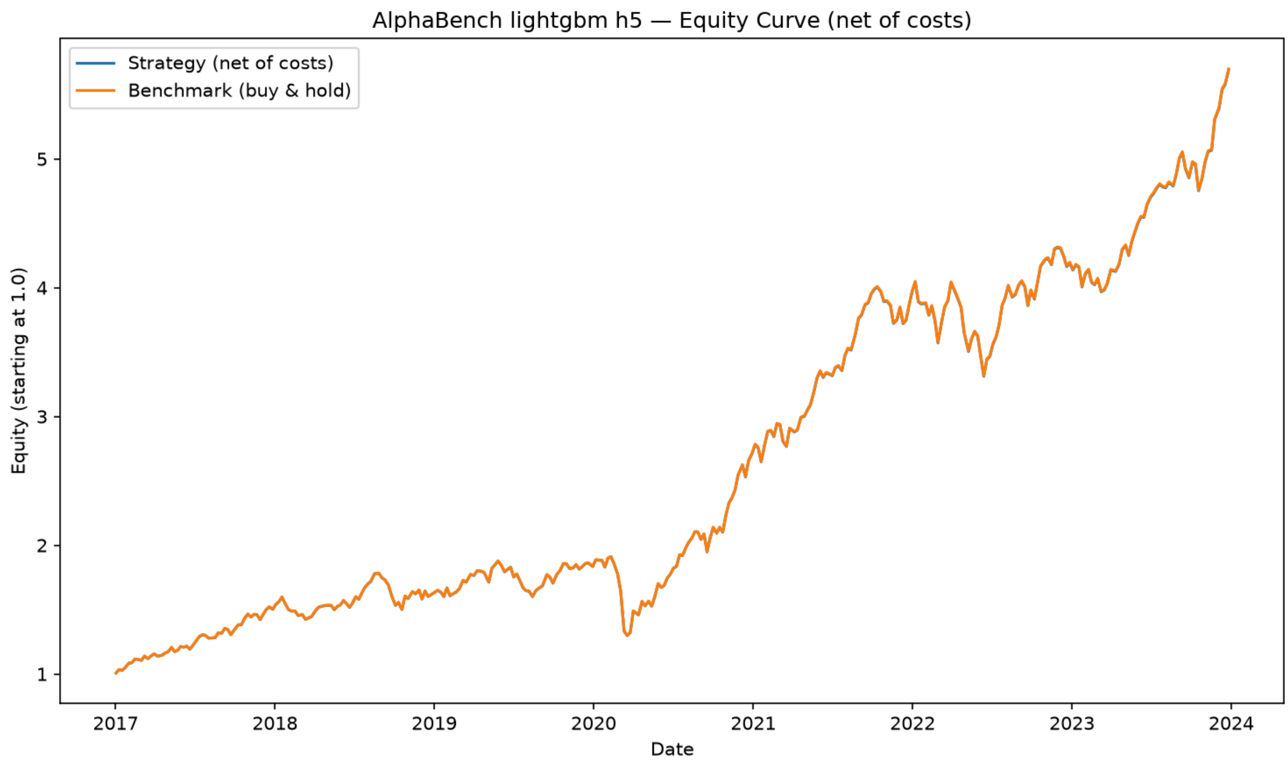
Backtest engine note. An h-day forward return makes consecutive rows' realised-return windows overlap by h−1 days; compounding overlapping windows at daily frequency multiplies the same underlying days' moves together several times over. `run_backtest` therefore restricts h>1 backtests to one shared, portfolio-wide rebalance date every h trading days before compounding — every symbol on the same dates, not each independently, which does not synchronise across symbols with different start dates. This was caught during development precisely because getting it wrong first produced an "annualised Sharpe of 3" and a ">700% single year", both obviously fabricated for a daily-equity long/flat strategy; the fix and both regression tests guarding it are in `evaluation/backtest.py` and `tests/test_backtest.py` .

With that fix, and NSE's h=1-calibrated `prob_threshold` (0.5164) applied unchanged to h=5 (consistent with "no tuning" — recalibrating it to h=5's own probability distribution specifically for this run would itself be a form of post-hoc tuning):

Metric	Strategy (h=5)	Buy-and-hold
Sharpe (annualised)	1.485	1.486
Ann. return	28.8%	28.9%
Excess Sharpe	**−0.001**	—

The strategy is statistically indistinguishable from buy-and-hold at this threshold — not because it found the same edge, but because the h=1-calibrated threshold is low relative to h=5's own probability distribution, so the model is long almost every period. `threshold_sensitivity` in `reports/metrics/backtest_results_h5.json` shows more differentiated (and still unremarkable) behaviour at higher quantile thresholds; this is reported as-is rather than re-thresholded to look more interesting, per the same no-post-hoc-tuning principle as everywhere else in this document.

Full results: `reports/metrics/walkforward_results_h5.json`, `reports/metrics/backtest_results_h5.json`, `reports/figures/equity_curve_h5.png`.



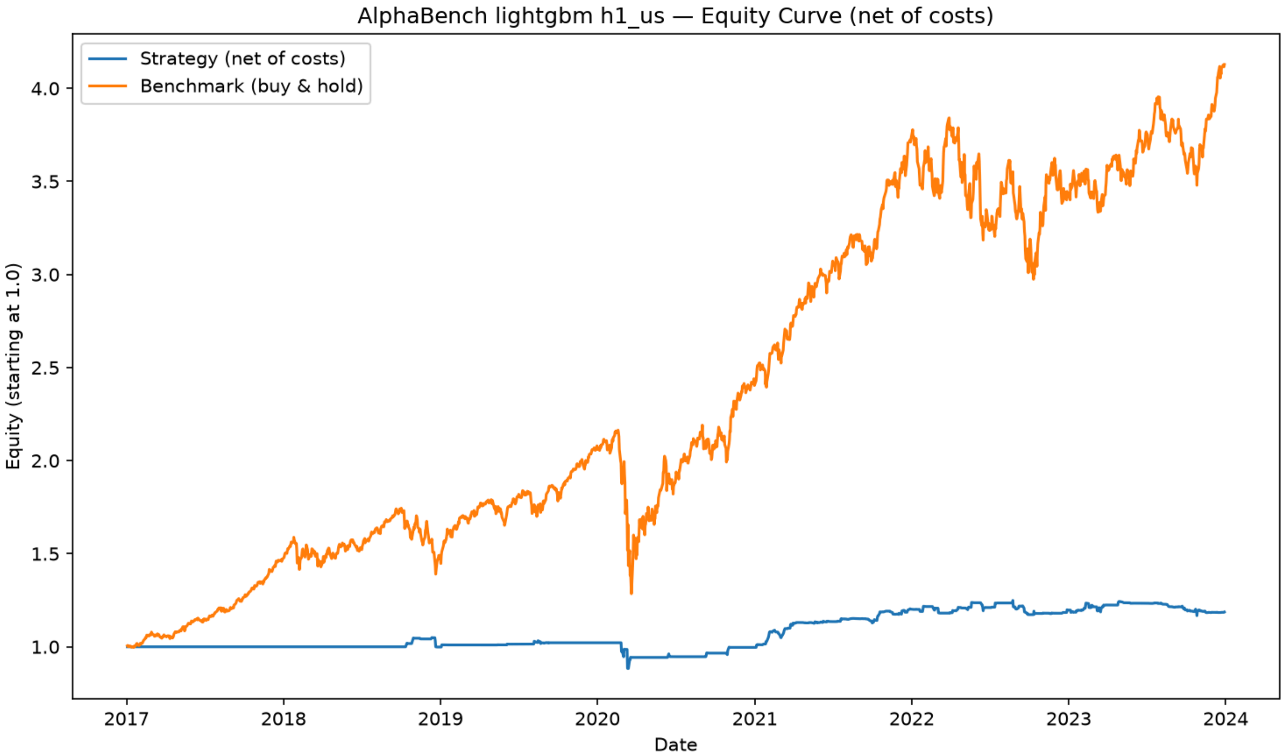
NSE h=5 equity curve, net of costs, against buy-and-hold. The two lines are visually indistinguishable: at the h=1-calibrated threshold the model holds a position almost every period, so the "strategy" is effectively buy-and-hold.

4. Week 9 generalisation test — a second, independent market

The identical M1 LightGBM h=1 pipeline (same `DEFAULT_PARAMS`, no retuning) was rerun on `config/universe_us.yaml` (30 US large-caps, benchmark SPY), in fully separate `data/*_us/` and `models/*_us/` paths so the NSE run is never touched:

Metric	NSE (primary)	US (generalisation)
Walk-forward AUC (mean, 7 folds)	0.506 ± 0.020	0.519 ± 0.028
Backtest Sharpe, net of 10 bps	0.108	0.349
Buy-and-hold Sharpe	1.797 (NIFTY)	1.053 (SPY)
Excess Sharpe	−1.691	−0.704
Bootstrap 95% CI, P(Sharpe>0)	[−0.45, 1.13], 0.699	[−0.22, 1.30], 0.884

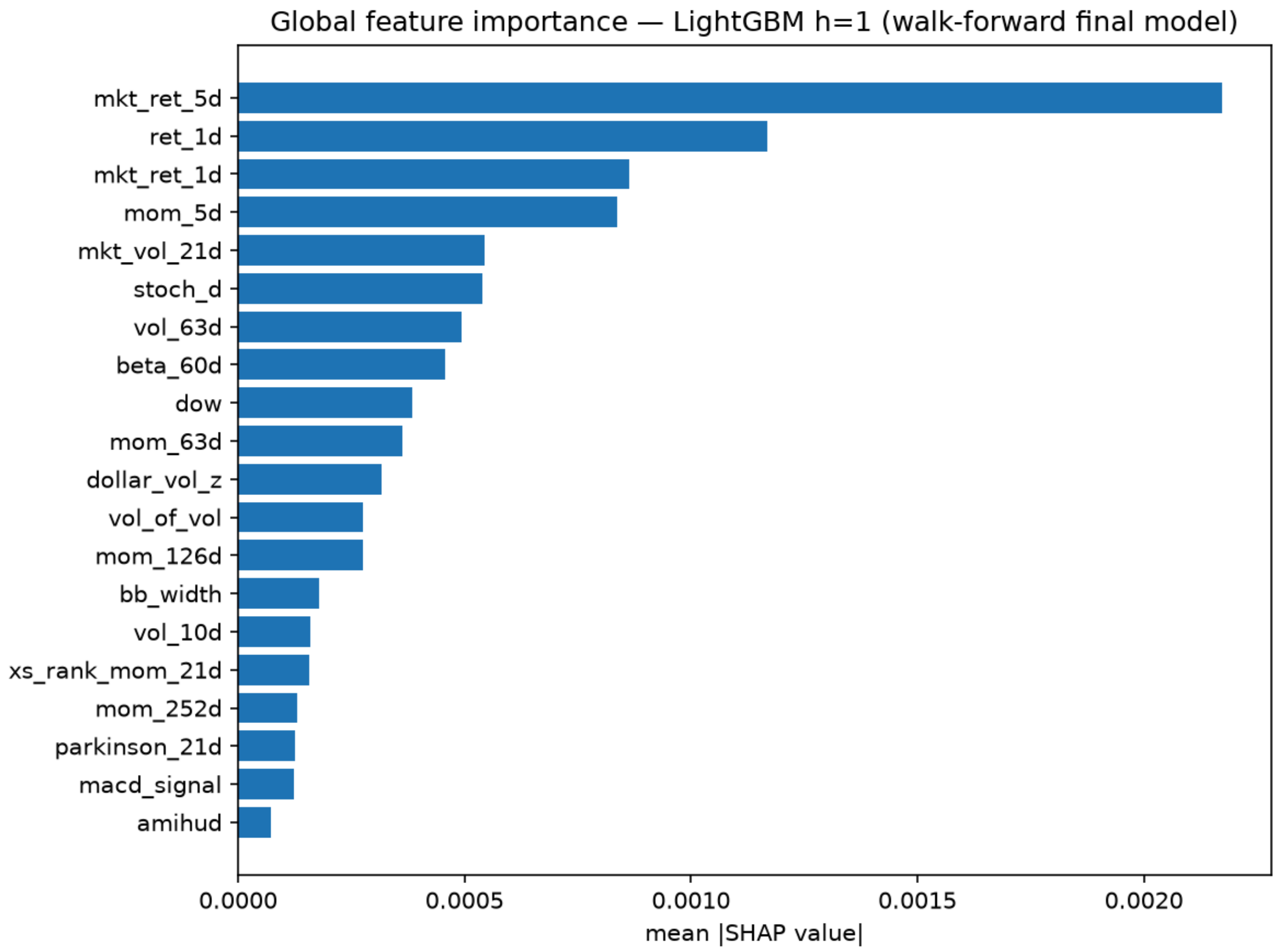
The US run shows a somewhat higher (still noise-level) AUC and a less negative excess Sharpe than NSE, but the qualitative finding is identical: a small, statistically unreliable edge that does not survive realistic costs against buy-and-hold. That the result replicates in direction and rough magnitude across two markets with different data-quality profiles, different benchmark dynamics, and no shared tuning is itself informative — it argues against "this particular NSE universe happened to be unlucky" and for "this feature set, at this data scale, does not carry a robust directional signal."



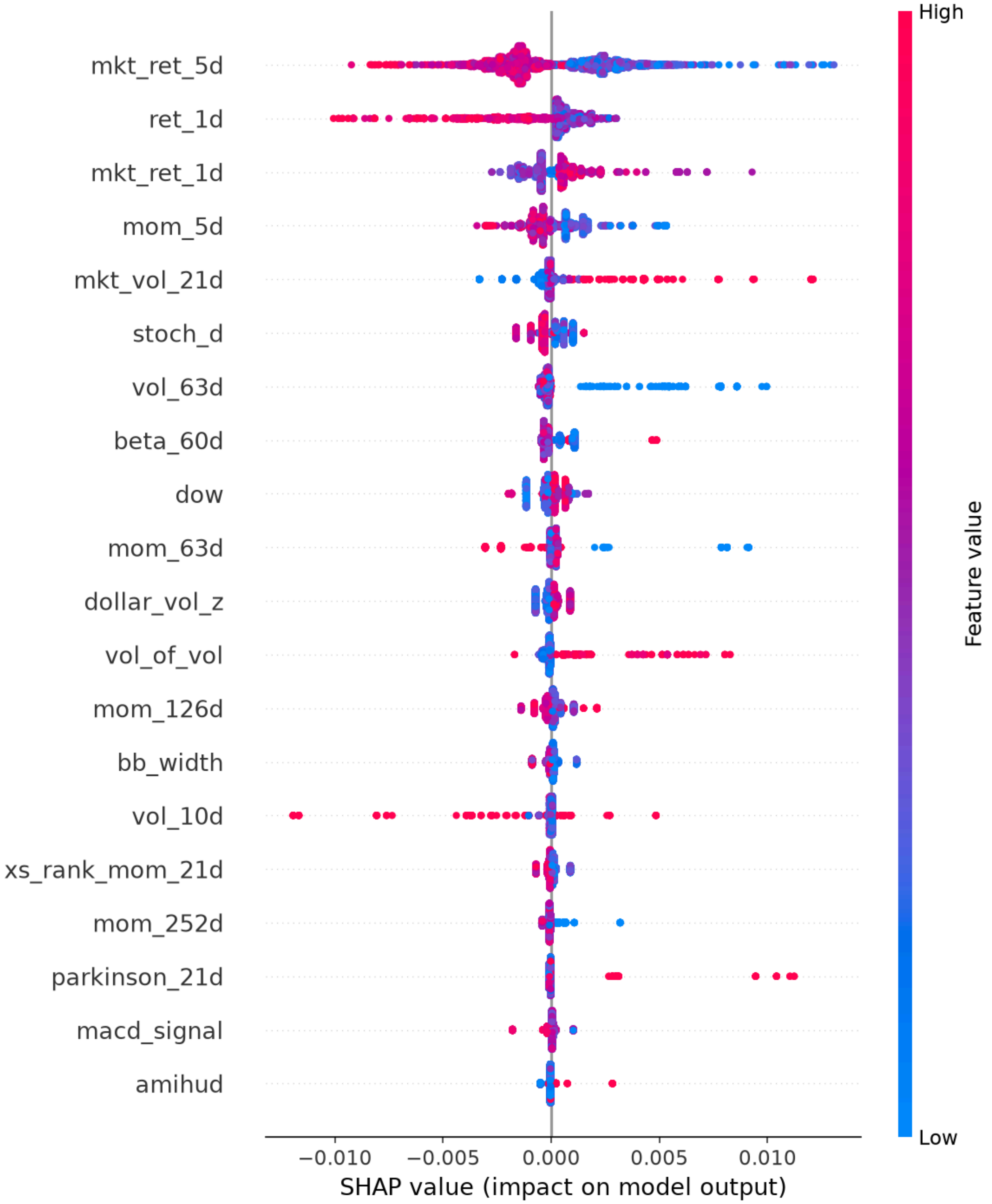
US h=1 equity curve, net of costs, against equal-weight buy-and-hold. Same shape as the NSE result: a strategy that trades, costs money, and trails the benchmark it is measured against.

5. Interpretability

SHAP (`TreeExplainer`) on the final LightGBM model, 5000-row sample of pre-holdout data (`reports/figures/shap_importance_h1.png` , `shap_beeswarm_h1.png`). The top features are market/momentum-driven — `mkt_ret_5d` , `ret_1d` , `mkt_ret_1d` , `mom_5d` , `mkt_vol_21d` — rather than idiosyncratic technical indicators, suggesting whatever the model is picking up leans toward broad market beta rather than stock-specific structure.



Global SHAP feature importance (mean |SHAP|) for the final LightGBM h=1 model.



SHAP beeswarm. Note the horizontal scale: individual contributions to the model's output are on the order of ±0.01 in probability terms — the model is barely moving off its base rate for any feature value.

Top-10-feature stability across the 7 folds' own models (`reports/metrics/shap_fold_stability_h1.json`): mean pairwise Jaccard overlap **0.352**. Moderate churn — a feature that matters in one fold is roughly as likely to drop out of the top 10 next fold as to stay — which is itself evidence against a stable, exploitable signal rather than for one. A genuinely robust edge should show more consistent feature importance across time than this.

6. The sealed holdout (SC-3, SC-4)

`reports/metrics/holdout_results.json`, written exactly once, by `alphabench evaluate-holdout`, which refuses to run a second time by design. Scored on `models/lightgbm_h1/final.joblib` (the `DEFAULT_PARAMS` model, never retrained on the Optuna search) against every row from **2025-01-01 onward** — 5,128 rows, 276 trading days, never touched by any fold, feature-selection decision, or tuning run before this.

Metric	Value
ROC-AUC	**0.5047**
95% bootstrap CI (block, 2000 resamples)	**[0.4817, 0.5266]**
Backtest Sharpe, net of 10 bps	**−0.583**
Buy-and-hold Sharpe (same window)	0.726
Excess Sharpe	−1.309
Ann. return (strategy / benchmark)	−1.3% / +10.2%

The CI straddles 0.50. The strategy loses money in absolute terms on the holdout while buy-and-hold gains 10%. This is the single most consequential number in the project, and it is fully consistent with every walk-forward fold, both markets, the deflated Sharpe, and the per-ticker dispersion reported above — nothing here comes as a surprise given the rest of the evidence.

7. Limitations

Survivorship bias (PROPOSAL section 5.4). The universe is today's 30 constituents, backfilled to 2010; companies delisted, acquired, or bankrupted over that window are absent, which biases the observed return distribution upward. This is not eliminated, only disclosed and bounded — point-in-time constituent membership is not available for free. Because the target here is relative direction rather than absolute return level, the effect on directional AUC is smaller than it would be on a long-only return backtest, but it is not zero, since any symbol's long-run drift being survivorship-conditioned still shapes the label distribution the model trains on.

Costs modelled, market impact not. 5+5 bps commission/slippage per side is applied uniformly; no model of price impact from the strategy's own trading, which at the position sizes implied by a 30-name equal-weight book is a reasonable simplification but would not scale to larger capital.

No point-in-time universe reconstruction; no intraday data; no leverage or shorting modelled beyond a flag. All explicitly out of scope per PROPOSAL section 3.2, not oversights.

Deflated Sharpe uses the Optuna search's trial count (50) applied to the reported backtest Sharpe, not a search directly over backtest configurations — a defensible but not perfectly literal application of Bailey & López de Prado's correction, since the Optuna objective was walk-forward AUC, not Sharpe. The shipped model was not selected by that search at all (see section 3), which is the more important guardrail: whatever the correction's precision, no configuration search informed which model's Sharpe is being reported as the headline number.

8. Conclusion

Across a full model ladder (B0 through M4), two independent markets, a 50-trial hyperparameter search with its selection bias corrected for, per-year and per-ticker breakdowns, and a sealed holdout touched exactly once, **AlphaBench finds no directional signal in daily equity returns that survives realistic transaction costs.** Walk-forward AUC sits at 0.49-0.52 for every model tried, with fold-to-fold noise as large as the gaps between models. The primary market's backtest underperforms simply buying and holding by a wide margin (excess Sharpe −1.69), the deflated Sharpe correction shows the modest observed edge doesn't clear the bar chance-plus-search-effort would produce on its own, and the sealed holdout — the number that matters most, precisely because it could not be p-hacked after the fact — lands at AUC 0.505 with a 95% CI straddling chance, and loses money net of costs while the benchmark gains 10%.

This is not a failed project. It is what a rigorous treatment of this question is expected to produce (PROPOSAL section 1, section 8.1): daily equity returns are close to a martingale difference sequence, and a result indistinguishable from chance, arrived at through a leakage-audited, walk-forward-validated, cost-aware pipeline with honest uncertainty bounds throughout, is the credible outcome — not evidence the pipeline is broken, and not a result to spin toward a more exciting conclusion than the data supports.